NETWORK RECONNAISSANCE & PORT SCANNING AUDIT

Complete Practical Guide + Illustrative Evidence Pack

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Lab target used in the illustrations: 192.168.56.102

Scanner illustration: Kali Linux — 192.168.56.101

IMPORTANT: The screenshots in this document are clearly marked SIMULATED EVIDENCE. They are examples for formatting and learning only. They are NOT proof that you personally ran these scans.
Replace them with screenshots from your authorized lab before submission.

1. What You Need to Complete

- TCP SYN scan (-sS)
- UDP scan (-sU)
- Service/version detection (-sV)
- OS detection (-O)
- Wireshark packet capture and analysis
- Open-port/service findings and security observations
- Final report PDF with real evidence
- One public link to the final PDF

Nmap's official documentation describes Nmap as a network exploration and security-auditing tool and documents SYN, UDP, service/version, OS detection, and output options. Use the commands below only against an authorized sandbox. cite turn0search3 turn0search4□ □ □ □

2. Lab IP Addresses — Example

Role	Example IP	Purpose
Kali Scanner	192.168.56.101	Runs Nmap/Wireshark
Target VM	192.168.56.102	Authorized vulnerable sandbox
Network	192.168.56.0/24	Isolated lab network

These addresses are deliberately chosen private lab examples. Use your real VM addresses in the actual assessment and report.

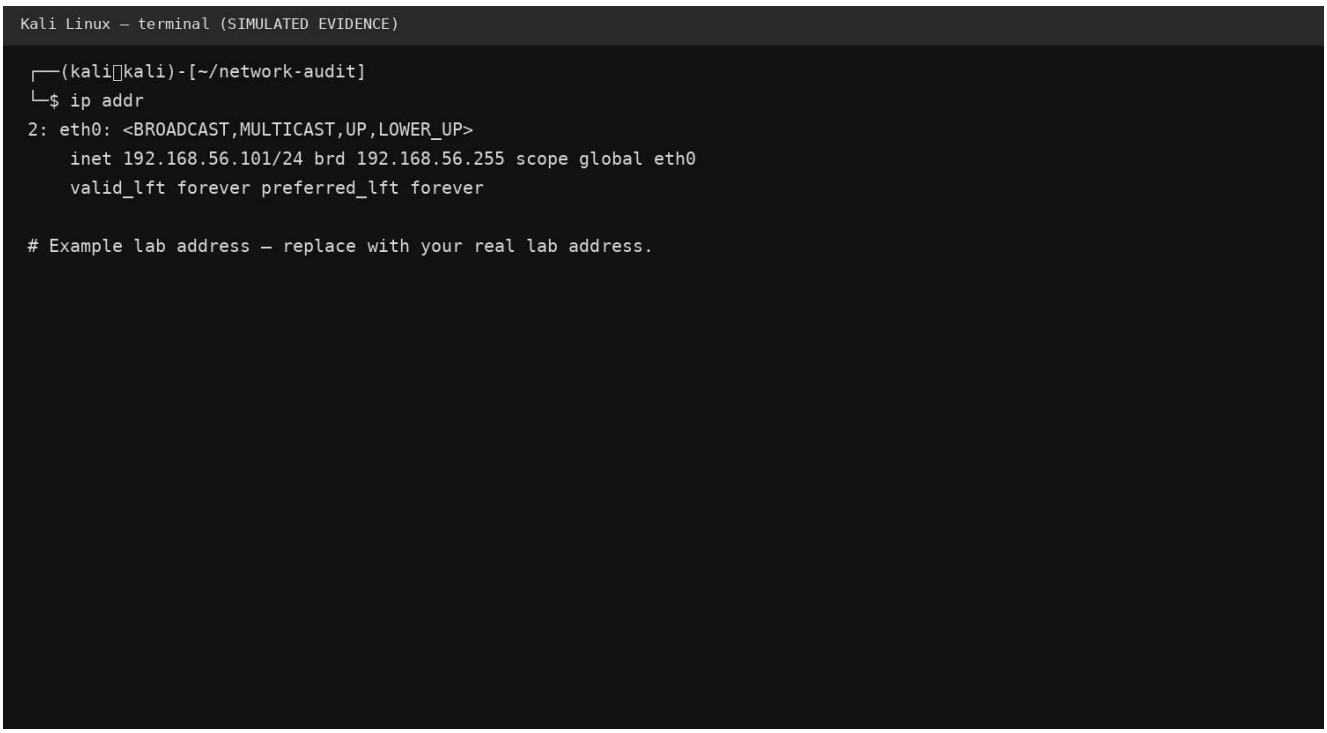


Figure 1 — SIMULATED: Kali IP address

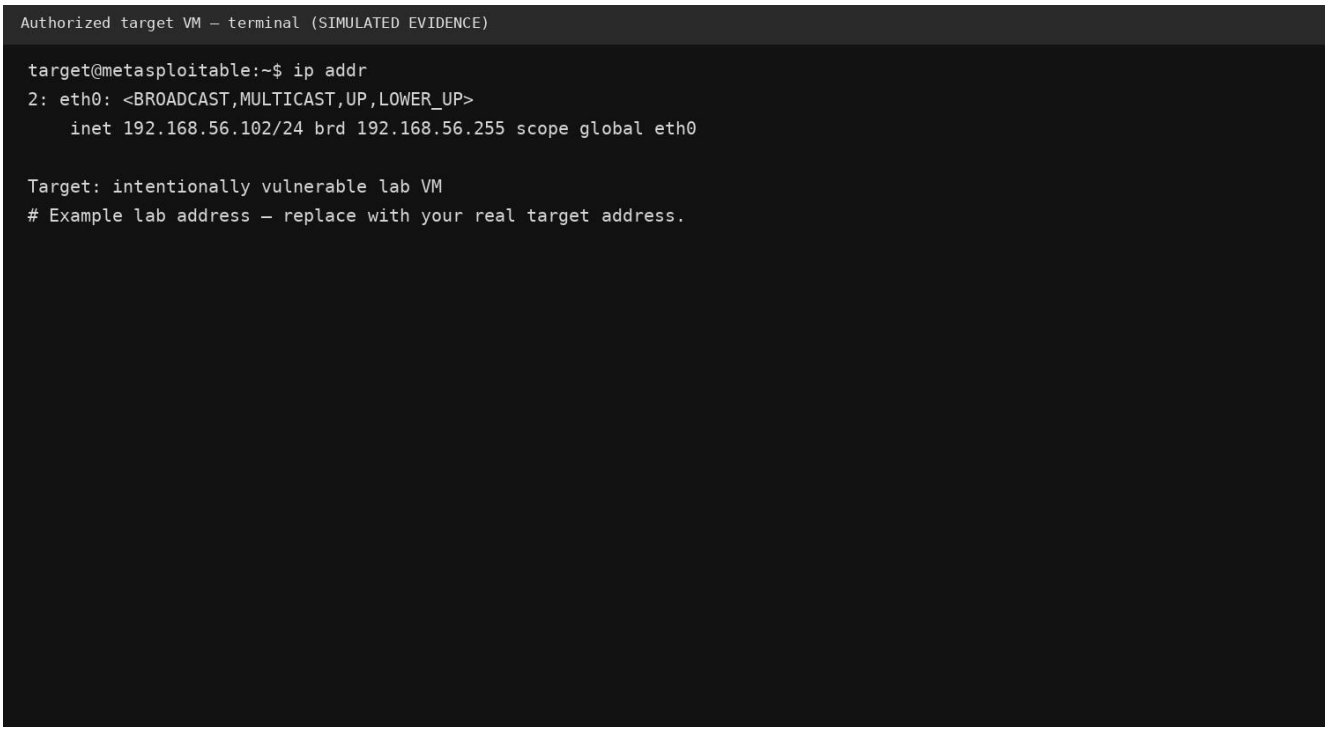


Figure 2 — SIMULATED: Target IP address

3. Verify Connectivity

ping -c 4 192.168.56.102

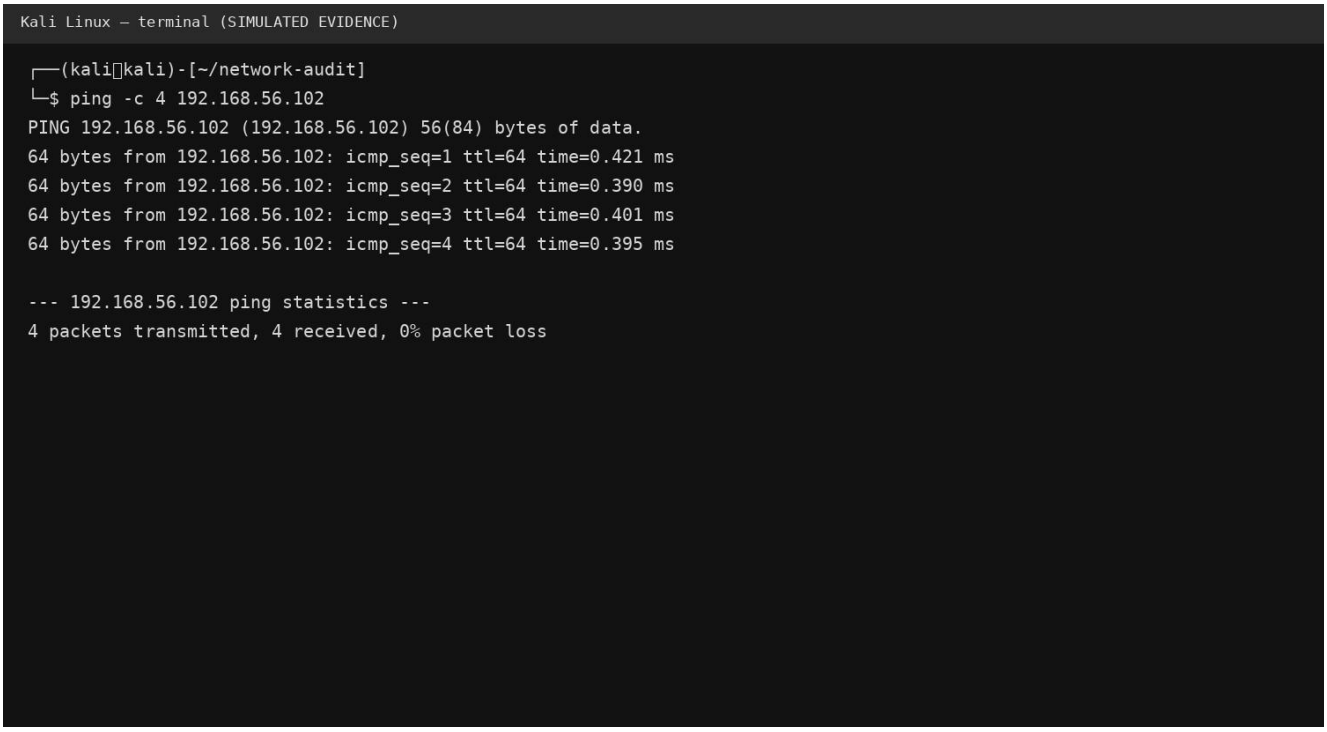


Figure 3 — SIMULATED: connectivity test

If your real target uses another address, replace 192.168.56.102. If ping fails, verify the isolated VM network and target IP before scanning.

4. Create Evidence Folder

mkdir -p ~/network-audit cd

~/network-audit

pwd

5. TCP SYN Scan

sudo nmap -sS -Pn 192.168.56.102 -oN tcp_syn_scan.txt

The official Nmap guide identifies -sS as the TCP SYN scan. The -oN option saves normal humanreadable output to a file. cite turn0search7 turn0search4

Kali Linux — Nmap (SIMULATED EVIDENCE)

```
$ sudo nmap -sS -Pn 192.168.56.102 -oN tcp_syn_scan.txt
Starting Nmap 7.95 ( https://nmap.org )
Nmap scan report for 192.168.56.102
Host is up (0.00058s latency).

PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
80/tcp   open  http
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned
```

Figure 4 — SIMULATED: TCP SYN scan

Replace the example target with your actual authorized target. Save your real screenshot as 04_tcp_syn_scan.png.

6. UDP Scan

sudo nmap -sU -Pn 192.168.56.102 -oN udp_scan.txt sudo nmap -sU --top-ports 100 -Pn 192.168.56.102 -oN udp_top100.txt

UDP scanning is generally slower than TCP scanning. Nmap documents -sU as its UDP scan option.

cite turn0search7

Kali Linux — Nmap UDP (SIMULATED EVIDENCE)

```
$ sudo nmap -sU --top-ports 100 -Pn 192.168.56.102 -oN udp_top100.txt
Starting Nmap 7.95 ( https://nmap.org )
Nmap scan report for 192.168.56.102
Host is up.

PORT     STATE         SERVICE
53/udp   open          domain
68/udp   open|filtered dhcpc

Nmap done: 1 IP address (1 host up) scanned
```

Figure 5 — SIMULATED: UDP scan

7. Service and Version Detection

sudo nmap -sV -Pn 192.168.56.102 -oN service_version_scan.txt

Kali Linux — Nmap service detection (SIMULATED EVIDENCE)

```
$ sudo nmap -sV -Pn 192.168.56.102 -oN service_version_scan.txt
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1
23/tcp   open  telnet      Linux telnetd
80/tcp   open  http        Apache httpd 2.2.8
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X
```

Figure 6 — SIMULATED: service/version detection

8. OS Detection

sudo nmap -O -Pn 192.168.56.102 -oN os_detection.txt

sudo nmap -O --osscan-guess -Pn 192.168.56.102 -oN os_detection.txt

Kali Linux — Nmap OS detection (SIMULATED EVIDENCE)

```
$ sudo nmap -O -Pn 192.168.56.102 -oN os_detection.txt
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.24 - 2.6.32

Warning: OSScan results may be unreliable because at least one open
and one closed port could not be found.
```

Figure 7 — SIMULATED: OS detection

9. Comprehensive Scan

sudo nmap -sS -sV -O -Pn 192.168.56.102 -oN comprehensive_scan.txt

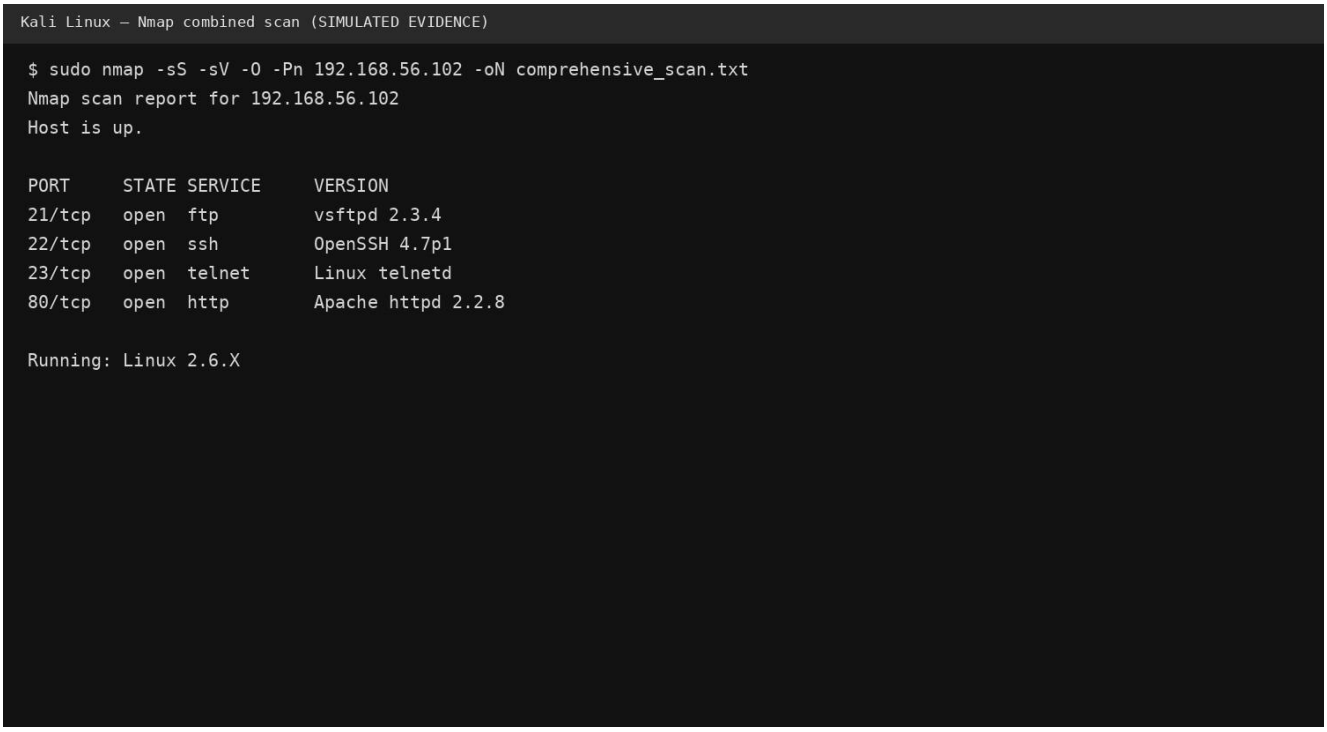


Figure 8 — SIMULATED: combined scan

10. Wireshark Capture

1. Open Wireshark with: wireshark
2. Select the interface connected to the isolated lab.
3. Start capture.
4. In a second terminal generate authorized traffic with ping and the Nmap scan.
5. Stop capture after the scan finishes.
6. Save as network_audit_capture.pcapng.

11. Wireshark Filters

- icmp — inspect ping traffic
- tcp — inspect TCP traffic
- http — inspect HTTP if your authorized lab generated it ftp — inspect FTP only if present in your authorized lab
- telnet — inspect Telnet only if present in your authorized lab

Wireshark — Authorized Lab Capture (SIMULATED EVIDENCE)

Display filter: icmp

No.	Time	Source	Destination	Protocol	Info
1	0.000	192.168.56.101	192.168.56.102	ICMP	Echo (ping) request
2	0.0004	192.168.56.102	192.168.56.101	ICMP	Echo (ping) reply
3	1.001	192.168.56.101	192.168.56.102	ICMP	Echo (ping) request

Packet Details

Internet Control Message Protocol (ICMP) — Echo request/reply observed during authorized connectivity testing.

Figure 9 — SIMULATED: ICMP packet analysis

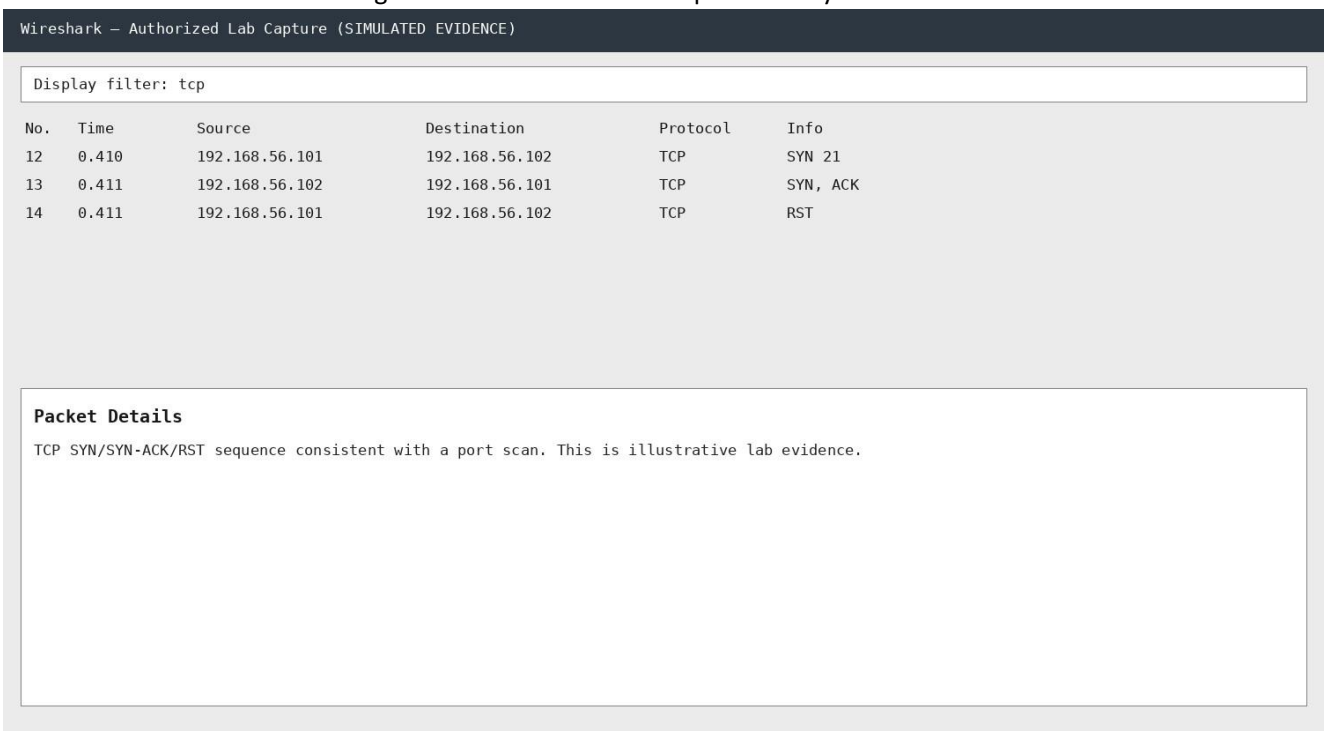


Figure 10 — SIMULATED: TCP packet analysis

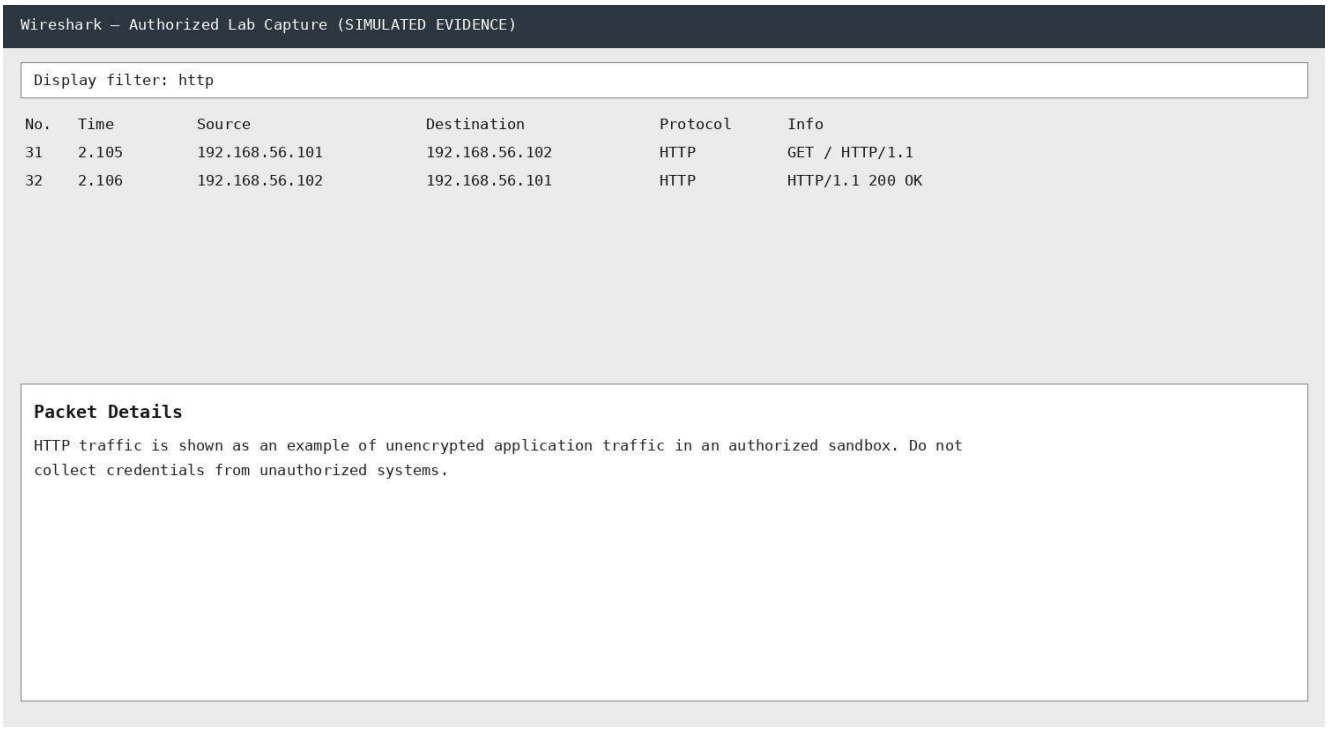


Figure 11 — SIMULATED: HTTP analysis example

12. What to Put in the Final Word Report

Executive Summary

Explain that the assessment was performed against an authorized simulated target using Nmap and Wireshark, and summarize the main observations.

Objective

Identify open TCP/UDP ports, services, versions, OS characteristics, and relevant packet-level observations.

Scope and Authorization

State the exact authorized target IP, testing machine, lab network, tools, and exclusions.

Methodology

Describe connectivity testing, TCP SYN scan, UDP scan, service/version detection, OS detection, and Wireshark capture.

Findings

Use only results actually observed in your scan and capture.

Recommendations

Suggest reducing unnecessary exposure, patching outdated software, using encrypted protocols, and restricting administrative services.

Conclusion

Summarize what the assessment demonstrated and how the target could be hardened.

13. Open Ports Table

Port	Protocol	State	Service	Version	Observation
21	TCP	Open	FTP	Actual result	Fill from Nmap
22	TCP	Open	SSH	Actual result	Fill from Nmap
23	TCP	Open	Telnet	Actual result	Fill from Nmap
80	TCP	Open	HTTP	Actual result	Fill from Nmap

The table above is a formatting example, not a claim about your target. Replace every value with your actual Nmap output.

14. Findings Template

Finding ID: F-01

Port/Protocol: _____21/TCP_____

Service/Version: _____vsftpd 2.3.4_____

Evidence: _____Nmap service/version scan identified TCP port 21 as open and running vsftpd 2.3.4._____

Risk: Low / Medium / High ---- Medium

Observation: _____An FTP service is exposed on the target. FTP is an older file-transfer protocol and, when used without additional protection, can transmit authentication and data without encryption._____

Recommendation: _____Disable FTP if it is not required. If file transfer is necessary, use a secure alternative such as SFTP/SSH or an appropriately secured FTPS configuration. Restrict access using firewall rules and strong authentication._____

15. Final Submission Process

7. Complete the real scans in your authorized lab.
8. Replace all SIMULATED screenshots with your own screenshots.
9. Check that the target IP in the report is your real target IP.
10. Insert the real Nmap outputs and Wireshark evidence.
11. Save the Word report.
12. Export it as Network_Reconnaissance_Findings_Report.pdf.
13. Upload the PDF to an allowed file-sharing service such as Google Drive.
14. Set General Access to Anyone with the link and Viewer.
15. Test the link in an Incognito/Private window.
16. Paste that ONE public link into Submit Deliverable.

16. Final Checklist

☑ Real target IP recorded

☐ TCP SYN scan completed

☐ UDP scan completed

☐ Service/version detection completed

☐ OS detection completed

☐ Wireshark capture completed

☐ Real screenshots inserted

☐ Open-port table completed

☐ Findings and recommendations completed

☑PDF exported

☑ Public link tested

☑ One link submitted

17. Accuracy & Ethics

- Use only an authorized sandbox.
- Do not scan public or third-party systems.
- Do not invent Nmap or Wireshark results.
- Do not present simulated screenshots as personal proof.
- If a protocol was not observed, say so.
- Do not publish credentials or sensitive information.

Nmap's own documentation emphasizes its use for network exploration and security auditing and provides a dedicated legal/appropriate-usage discussion. Keep this assignment inside the authorized lab boundary. cite turn0search1 turn0search2